# F70LB Life Insurance Mathematics B

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## 10 Single Figure Indices

How do we compare the mortality experience of two or more populations?

A single figure that summarises the mortality experience of a population enables easy comparison.

Single-figure indices are formulas for calculating such a single figure:

Notes on single-figure indices:

- they are easy to assimilate, rather than a range of age-specific rates; but
- there is a loss of information in summarising and so care needs to be taken in the construction and application of the indices

### 10.1 Data for examples on single figure indices

Table 1: Mortality statistics

| Age Group | Region A population | Region A deaths | Region B population | Region B deaths | Country population | Country deaths |
|---|---|---|---|---|---|---|
| 0–9 | 2000 | 20 | 1000 | 10 | 55000 | 530 |
| 10–19 | 1500 | 5 | 1000 | 3 | 50000 | 160 |
| 20-29 | 1000 | 2 | 1000 | 2 | 60000 | 120 |
| 30–39 | 800 | 2 | 950 | 2 | 45000 | 95 |
| 40–49 | 700 | 2 | 900 | 2 | 45000 | 105 |
| 50–59 | 600 | 4 | 800 | 5 | 40000 | 250 |
| 60–69 | 500 | 10 | 800 | 16 | 30000 | 550 |
| 70–79 | 400 | 30 | 700 | 52 | 20000 | 1400 |
| 80–89 | 300 | 50 | 600 | 100 | 10000 | 1700 |
| 90+ | 100 | 50 | 150 | 75 | 2000 | 1000 |
| Totals | 7900 | 175 | 7900 | 267 | 357000 | 5910 |

Table 2: Age-specific mortality rates

| Age group | Region A | Region B | Country |
|---|---|---|---|
| 0–9 | 0.010 | 0.010 | 0.010 |
| 10–19 | 0.003 | 0.003 | 0.003 |
| 20–29 | 0.002 | 0.002 | 0.002 |
| 30–39 | 0.003 | 0.002 | 0.002 |
| 40–49 | 0.003 | 0.002 | 0.002 |
| 50–59 | 0.007 | 0.006 | 0.006 |
| 60–69 | 0.020 | 0.020 | 0.018 |
| 70–79 | 0.075 | 0.074 | 0.070 |
| 80–89 | 0.167 | 0.167 | 0.170 |
| 90+ | 0.500 | 0.500 | 0.500 |

## 10.2 Crude Death Rate (CDR)

$$\begin{aligned} \text{CDR} &= \frac{\text{Total deaths}}{\text{Total exposure}} \\ &= \frac{\sum_x d_x}{\sum_x \mathrm{E}_x^c} \end{aligned}$$

Where:

$$\begin{aligned} d_x &= \text{number of deaths in age group } x \\ \mathrm{E}_x^c &= \text{“population in age group } x\text{”} \end{aligned}$$

In the handout example we have:

$$\begin{aligned} \text{Region A:} \quad & \text{CDR} = \frac{175}{7,900} = 0.022 \\ \text{Region B:} \quad & \text{CDR} = \frac{267}{7,900} = 0.034 \\ \text{Country:} \quad & \text{CDR} = \frac{5,910}{357,000} = 0.017 \end{aligned}$$

Notes:

1. $$\begin{aligned} \text{CDR} &= \frac{\sum_x d_x}{\sum_x \mathrm{E}_x^c} = \frac{\sum_x \mathrm{E}_x^c \frac{d_x}{\mathrm{E}_x^c}}{\text{Total population}} \\ &= \sum_x \left( \frac{\mathrm{E}_x^c}{\text{Total population}} \right) \mu_x \\ &= \sum_x w_x \, \mu_x \end{aligned}$$

   Hence, the CDR is a weighted average of the age-specific forces of mortality, where the weights reflect the population structure.

2. Strengths of the CDR:
   - CDR is easy to calculate
   - the data requirements are minimal (just the total deaths and the total population).

3. Weaknesses of the CDR:
   - CDR is heavily influenced by the age structure of the population.

In the handout example the CDRs imply that:

- Region B has **heavier mortality than Region A**
- Both regions have **heavier mortality than the country**

However, the age specific mortality indicate that:

$$\text{Mortality Region A} \approx \text{Mortality Region B} \approx \text{Mortality of Country}$$

CDR for Region A is influenced by the relatively high proportion of young people.

The country has, relatively, an even younger population than either Region A or B.

We need to develop an index which is less influenced by differences in age structures.

## 10.3 Standardised Death Rate (SDR)

$$\text{SDR} = \sum_x w_x \, \mu_x \qquad \text{(never SMR!)}$$

$$\text{Where} \qquad w_x = \frac{{}^s\mathrm{E}_x^c}{\sum_x {}^s\mathrm{E}_x^c} \text{ are based on a standard population}$$

We note that:

1. For Region A, (with age specific ${}^A\mu_x$):

$$\text{SDR}_A = \frac{\sum_x {}^s\mathrm{E}_x^c \, {}^A\mu_x}{\sum_x {}^s\mathrm{E}_x^c}$$

   In the handout example (taking the country's population structure as standard), we have for $SDR_A$:

$$\frac{\frac{20}{2{,}000}55{,}000 + \frac{5}{1{,}500}50{,}000 + \cdots + \frac{50}{100}2{,}000}{55{,}000 + 50{,}000 + \cdots + 2{,}000} = 0.0171$$

   And $\text{SDR}_B = 0.01689$

2. The SDR is *not* very influenced by the population structure as shown by:

$$\mathbf{SDR}_A \approx \mathbf{SDR}_B$$

3. The data requirements are:

   - age specific mortality rates of the region
   - age specific structure of the standard population (weights)

   However, the weights (age specific structure of the standard population) may be unknown or unreliably estimated.

   We consider the case where the weights are only reliably known at specific times (eg census times).

### 10.3.1 Indirectly Standardised Death Rate

The calculation of the SDR is an example of direct standardisation.

However the age specific structure of a standard population may only be known at census dates.

This gives problems if we need to calculate the SDR at dates in between these censuses.

**Assumption:**

$$\frac{\textbf{SDR (at census)}}{\textbf{CDR (at census)}} = \frac{\textbf{SDR (non-census)}}{\textbf{CDR (non-census)}}$$

Then we have that:

$$\begin{aligned} \text{SDR (non-census)} &= \frac{\text{SDR (at census)}}{\text{CDR (at census)}} \\ &\quad \times \text{CDR (non-census)} \\ &= F \times \text{CDR (non-census)} \end{aligned}$$

and the data requirements for CDR (non-census) are easily met even at non-census times.
$F$ is called the **area comparability factor, and:**

$$F = \frac{\frac{\sum_x {}^s\mathrm{E}^c_x \mu_x}{\sum_x {}^s\mathrm{E}^c_x}}{\frac{\sum_x d_x}{\sum_x \mathrm{E}^c_x}}$$

$F$ is calculated at the census and assumed to remain constant until the next census when it is updated.

## 10.4 Standardised Mortality Ratio

The SMR is a ratio (not rate) of:

$$\frac{\text{actual deaths in region}}{\text{expected deaths in region}}$$

The expected deaths are calculated using the **age-specific mortality rates for the standard population.**

Therefore, the SMR for region $A$ relative to a standard population $S$, is:

$$\text{SMR}_A = \frac{\sum_x {}^A\mu_x \, {}^A\text{E}_x^c}{\sum_x {}^s\mu_x \, {}^A\text{E}_x^c}$$

Notes:

1. SMR is the most common ratio found in demographic studies.
2. In our example, $\text{SMR}_A$:

$$\frac{20 + 5 + \cdots + 50}{2,000\frac{530}{55,000} + 1,500\frac{160}{50,000} + \cdots + 100\frac{1,000}{2,000}} = 1.022 (= 102.2\%)$$

   $\mathbf{SMR}_B = 1.009 \quad (= 100.9\%)$

3. The data requirements are:
   - total number of deaths in the region
   - population at each age group in the region
   - age-specific mortality rates in standard population

   This information is more likely to be known more reliably than the information for the SDR (in particular the age specific population structure).

4. The SMR is usually expressed as a percentage and:
   - a region which has the same mortality as the standard population will have an SMR of 100%
   - if the SMR is greater than 100% then the region has heavier mortality than the standard population.
   - if the SMR is less than 100% then the region has lighter mortality than the standard population.